

Sr. No.	Oral Examination Questions
1.	What is a computer network?
2.	What are the different types of Networks?
3.	What are LAN, MAN, and WAN?
4.	What is a protocol?
5.	What is network topology?
6.	What are the different types of topologies?
7.	What are the advantages and disadvantages of a star network topology?
8.	What are the advantages and disadvantages of a bus network topology?
9.	Can you describe a ring network topology and its characteristics?
10.	For six devices in a network, what is the number of cable links required for a mesh, ring, bus, and star topology?
11.	What are the different types of transmission medium?
12.	What are some examples of guided transmission media?
13.	Give examples of unguided transmission media.
14.	What is the difference between client-server, peer-to-peer, and hybrid architecture?
15.	What are different network devices?
16.	What is a router, switch, bridge, and access point?
17.	How many layers are there in OSI and TCP/IP model?
18.	What is the function of each layer? Physical layer Datalink layer Network layer Transport layer Session layer Presentation layer Application layer
19.	What is the unit of communication at each layer?
20.	What is the function of Data-Link-Layer?
21.	What is, Flow Control Error Control Congestion Control
22.	What are design issues of Data-Link Layer?
23.	What are different techniques to framing?
24.	What is CRC?
25.	What is Hamming Code?
26.	What are flow control protocols? Simplex Protocol Stop and Wait Protocol

	Sliding-Window Protocol Go-back-N and Selective-Repeat Protocol
27.	What are Multiple Access Protocols: Pure and Slotted ALOHA, CSMA, WDMA, CSMA/CD, CSMA/CA?
28.	What is the function of the Network Layer?
29.	What are different network layer protocols?
30.	What is IP protocol?
31.	What is an IP address?
32.	What is the length of the IP address?
33.	What is the address space of IPv4?
34.	What are the different classes of IP addresses?
35.	Which address comes in Class-A, B, C, D, and E?
36.	What is NAT (Network Address Translation)?
37.	What is Subnetting and supernetting?
38.	What is ARP, RARP, ICMP?
39.	What is routing?
40.	What are different routing algorithms and protocols?
41.	What is distance-vector routing?
42.	What is link-state routing?
43.	What is path-vector routing?
44.	Routing Protocols: RIP, OSPF, BGP?
45.	What is the function of Transport-Layer?
46.	What is process-to-process communication?
47.	What is port-number?
48.	What are the different categories of port-numbers?
49.	What are well-known port number?
50.	What are private/ephemeral/dynamic port number?
51.	What are registered ports?
52.	What are different transport layer protocols?
53.	What is UDP?
54.	What services are provided by UDP?
55.	What is TCP?
56.	What services are provided by TCP?
57.	What is Flow control, error control, congestion control?
58.	What connection in TCP?
59.	What is a three-way-handshaking connection establishment?
60.	What is a three-way-handshaking connection termination?
61.	What is SCTP?
62.	What is the function of application layer?
63.	What are different application layer protocols?
64.	What is role of protocol DHCP?
65.	What is role of protocol DNS?

66.	What is role of protocol TELNET?
67.	What is role of protocol FTP?
68.	What is role of protocol HTTP?
69.	What is role of protocol SMPT, POP3, IMAP4 and MIME?
70.	What is role of protocol SNMP?
71.	What are security goals?
72.	What is confidentiality, integrity and availability?
73.	What are different security attacks?
74.	What is symmetric-key cryptography?
75.	What is asymmetric-key cryptography?
76.	What is difference between symmetric-key cryptography and asymmetric-key cryptography?
77.	How does IPSec provide security at the network layer?
78.	What is the purpose of a firewall, and how does it protect a network?
79.	What is SSL and HTTPS?
80.	Comment about Security in Network, Transport and Application?